Chapter 4, Section 3

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1. If X is a curve of genus 2, show that a divisor D is very ample $\iff$ $\deg D \geq 5$. This strengthens (3.3.4).

Proof. Let D be a very ample divisor on X. By the degree-genus formula, a nonsingular plane curve cannot have genus equal to 2, which implies $\ell(D) \geq 4$. By the Riemann-Roch theorem, we have

$$\deg D = \ell(D) - \ell(K - D) + 1 \geq 5 - \ell(K - D),$$

so it suffices to show $\ell(K - D) = 0$. Indeed, $\deg D \geq 3$ by (Ex. 1.5), so $K - D$ has negative degree. Hence, $\ell(K - D) = 0$. $\square$

3. If X is a curve of genus ≥ 2 which is a complete intersection (II, Ex. 8.4) in some $\mathbf{P}^n$, show that the canonical divisor K is very ample. Conclude that a curve of genus 2 can never be a complete intersection in any $\mathbf{P}^n$.

Proof. Let d be the degree of X in $\mathbf{P}^n$. Then X is a complete intersection of $n-1$ hypersurfaces of degree d_i, and by (II, Ex. 8.4), we have $\mathcal{O}_X(K) \cong \mathcal{O}_X(e)$, where $e = \sum d_i - n - 1$, so it suffices to show $e \geq 1$. Indeed, $\deg K = ed$ and $\deg K \geq 2$. Hence, $e \geq 1$.

If X is a curve of genus 2, then $\ell(K) = 2$, so K corresponds to a morphism $X \to \mathbf{P}^1$, which can never be an embedding. $\square$

6. *Curves of Degree 4.*

(a) If X is a curve of degree 4 in some $\mathbf{P}^n$, show that either

(1) $g = 0$, in which case X is either the rational normal quartic in $\mathbf{P}^4$ (Ex. 3.4) or the rational quartic curve in $\mathbf{P}^3$ (II, 7.8.6), or

(2) $X \subseteq \mathbf{P}^2$, in which case $g = 3$, or

(3) $X \subseteq \mathbf{P}^3$ and $g = 1$.

(b) In the case $g = 1$, show that X is a complete intersection of two irreducible quadric surfaces in $\mathbf{P}^3$ (I, Ex. 5.1).

Proof.

(a) $\boxed{n = 2}$ The degree-genus formula gives

$$g = \frac{(d-1)(d-2)}{2} = 3.$$

$\boxed{n = 3}$ If X is not contained in any hyperplane, then $g < 3$ (III, Ex. 9.11) (Ex. 3.5b). Since $g \neq 2$ by (Ex. 3.1), X is either the rational quartic curve in $\mathbf{P}^3$, or an elliptic curve.

$\boxed{n = 4}$ X is rational by (Ex. 1.5).

(b) Let $\mathcal{I}$ be the ideal sheaf of X in $\mathbf{P}^3$, and consider the exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbf{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbf{P}_3}(2)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbf{P}^3, \mathcal{I}(2)) \longrightarrow \mathrm{H}^0(\mathbf{P}^3, \mathcal{O}_{\mathbf{P}^3}(2)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

Now, X is an elliptic curve, and $\mathcal{O}_X(2)$ corresponds to a divisor D of degree 8. The Riemann Roch theorem gives $\dim \mathrm{H}^0(X, \mathcal{O}_X(2)) = 8$, and the vector space in the middle has dimension 10, so we conclude that $h^0(\mathbf{P}^3, \mathcal{I}(2)) \geq 2$. An element of $\mathrm{H}^0(\mathbf{P}^3, \mathcal{I}(2))$ is a form of degree 2, whose zero-set is a quadric surface containing X. Any such quadric must be irreducible and reduced because X is not contained in any hyperplane. Hence, X is contained in two distinct irreducible quadric surfaces and is a complete intersection because X is a curve of degree 4.

□

9. Prove the following lemma of Bertini: if X is a curve of degree d in $\mathbf{P}^3$, not contained in any plane, then for almost all planes $H \subseteq \mathbf{P}^3$ (meaning a Zariski open subset of the dual projective space $(\mathbf{P}^3)^*$), the intersection $X \cap H$ consists of exactly d distinct points, no three of which are collinear.

Proof. □

12. For each value of $d = 2, 3, 4, 5$ and r satisfying $0 \leq r \leq \frac{1}{2}(d-1)(d-2)$, show that there exists an irreducible plane curve of degree d with r nodes and no other singularities.

Proof. There is nothing to prove for $d = 2$ or $r = 0$, so assume otherwise. By (Ex. 1.8) (3.10), for each $d = 2, 3, 4, 5$, we want to show there exists a smooth curve of degree d and genus $0 \leq g \leq \frac{1}{2}(d-1)(d-2)$ in some $\mathbf{P}^n$.

$\boxed{d = 2}$ There is nothing to prove.

$\boxed{d = 3}$ $g = 0$ corresponds to the rational cubic in $\mathbf{P}^3$, $g = 1$ □